SULIT

CP115
Computer Programming I
Semester I
Session 2026/2027
2 Hours 30 Minutes

CP115
Pengaturcaraan Komputer I
Semester I
Sesi 2026/2027
2 Jam 30 Minit

Nama	No Matrik	Kelas

KOLEJ MATRIKULASI LABUAN
LABUAN MATRICULATION LABUAN

PRA-PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI
MATRICULATION PROGRAMME PRE-EXAMINATION

PENGATURCARAAN KOMPUTER I
2 JAM 30 MINIT

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU.
DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.

Untuk Kegunaan Pemeriksa		
No Soalan	Markah	
	Pemeriksa	KP/KKP
1 (17)		
2 (15)		
3 (17)		
4 (19)		
5 (7)		
6 (10)		
Jumlah (85)		

Kertas soalan ini mengandungi 10 halaman bercetak
This question paper consists of 10 printed pages

1.

a. State and define the second and third steps in problem solving.

[4 marks]

b. For each situation below, identify whether each is Input, Process or Output.

[5 marks]

Situations	Answer
A barcode scanner reads the code on a product at the checkout counter.	
A fitness watch converts the number of steps taken into calories burned.	
A printer produces a monthly payslip for an employee.	
A user speaks a voice command into a smart speaker.	
A navigation app shows the fastest route on the phone screen.	

c. For each situation below, identify the Input, Process and Output.

i. A car park system determines whether a vehicle is allowed to enter a covered car park. The system receives the vehicle's height in metres as input. It checks if the height is 2.1 or below. If the height is 2.1 or below, it outputs "ALLOWED". If the height is above 2.1, it outputs "NOT ALLOWED".

[4 marks]

Input	
Process	
Output	

ii. Every month, Kumar needs to calculate the monthly salary for each of his 8 workers at his car wash in Labuan. For each worker, he has to enter the number of hours worked and the hourly pay rate.

[4 marks]

Input	
Process	
Output	

2.

a. An algorithm can be represented in many ways including pseudocode and flowchart. Define the meaning of pseudocode.

[1 mark]

b. Draw the shape of the flowchart components based on the description.

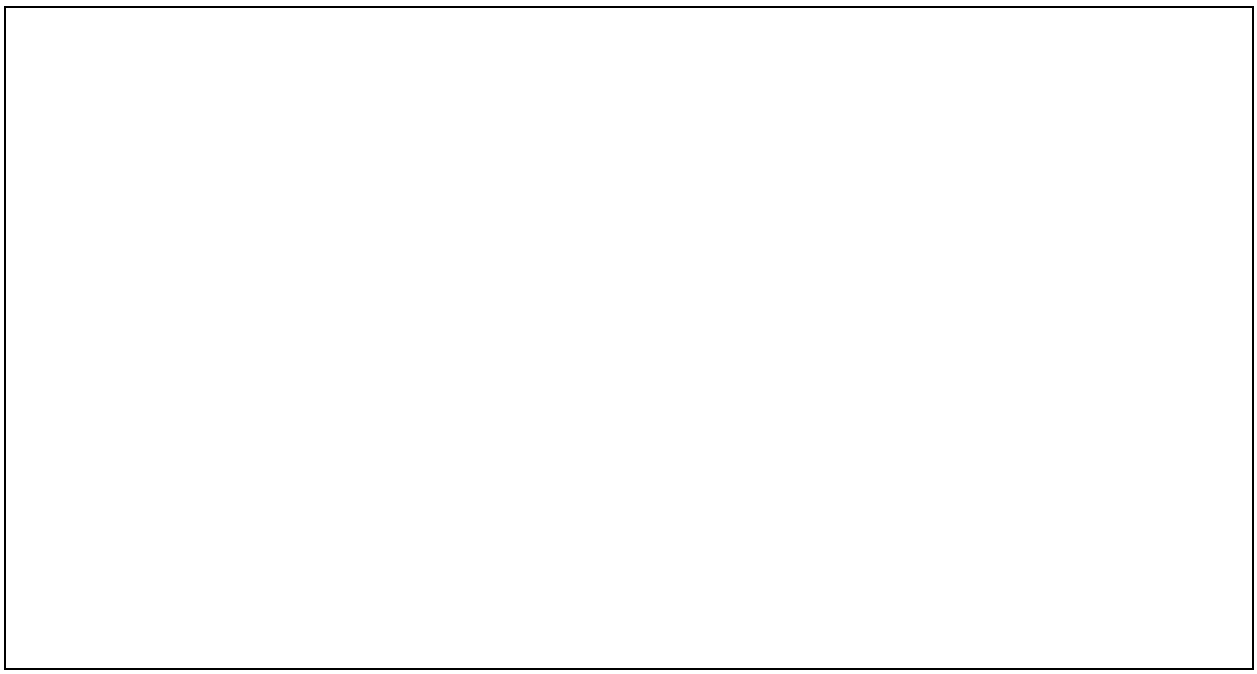
[3 marks]

Description	Answer
Used to indicate the beginning or the end of a flowchart	
Used to show the direction of the flow of the process	
Used to define a decision	

c. Convert the pseudocode below into a flowchart.

[5 marks]

```
START
    Enter temperature
    if temperature > 37.5
        Print "You have a fever"
    else
        Print "Your temperature is normal"
    end if
END
```



d. Encik Rahman runs a small cinema and needs to calculate the total number of tickets sold in a day. Write a pseudocode that processes ticket sales for 15 customers throughout the day. For each customer, the system gets the number of tickets bought and adds it to the total tickets sold. After serving all 15 customers, display the total number of tickets sold.

[6 marks]

3.

a. A program written in a high-level language must be translated before it can be executed by the computer. Define what a compiler is.

[1 mark]

b. Give one (1) advantage and one (1) disadvantage of using high-level language.

[2 marks]

Advantage	Disadvantage

c. Determine whether each of the following identifiers is valid or invalid.

[4 marks]

Identifier	Answer
total_mark	
2ndPlace	
price$	
_Count9	

d. For the Python code below, a user executes the program and enters Ali as the input. Identify which line will throw an error and what type of error it is.

[2 marks]

```
1. name = input("Enter your name")
2. age = 17
3. message = name + " is " + age + " years old"
4. print(message)
```

Line of code	Type of error

e. State the output of the Python program below.

[4 marks]

```
1. word = "Labuan Matrik"
2. print(len(word))
3. print(len(word) % 5)
4. print(word.lower())
5. print(f"First letter is {word[0]}")
```

f. State two (2) ways of writing comments in Python.

[2 marks]

g. Convert the following arithmetic expressions into Python statements.

[2 marks]

Arithmetic expression	Python statements
$p = \frac{(a+b)^2}{2c}$	
$q = \sqrt{x^2 + y^2} - 3x$	

4.

a. Compare the usage of the below Python keywords for repetition control structures.

[2 marks]

for	***while***

b. Mei Ling has written the Python code below that allows her to enter the prices of items she buys and the program will add up the total. The program will continually run until she enters 0. Identify one (1) component of the sentinel loop that she did not add to her code.

[1 mark]

```
1. total = 0
2. price = float(input("Enter the price"))
3. while price != 0 :
4.     total = total + price
5. print("Total price is ", total)
```

c. Convert the below Python statements using *for* statements.

[4 marks]

```
1. i = 3
2. while i <= 30 :
3.     print(i)
4.     i = i + 3
5. print("Done")
```


d. Write a code segment for each of the following selection control structures based on the given situation.

[3 marks]

Type of Selection Control Structure	Code Segment
Single selection Display "Free delivery" if the variable *purchase* is more than 100.	
Dual selection Display "Even" if the variable *num* is an even number, otherwise display "Odd".	
Multiple selection Display "A" if the variable *mark* is 80 or above, "B" if 60 or above, otherwise display "C".	

e. Evaluate the Python statements below and find the value of each variable.

[4 marks]

Python statements	Answer
y = 17 % 5 == 2 and 9 < 4	y =
x = not (7 > 2) or 3 != 3	x =
z = (20 + 4) / 6 == 4.0	z =
a = 9 // 2 == 4 or 1 > 5	a =

f. Convert the below pseudocode into a Python program.

[5 marks]

```
1. START
2.  n = 20
3.  While n > 0
4.      If n % 4 == 0
5.          Print n
6.      end if
7.      n = n - 2
8.  end while
9. END
```

5. Write a Python program that will continually accept the weight of parcels in kilograms ranging from 1 - 30. The program will terminate if the user enters any value outside of that range. Assume all the input the user will enter is only numbers. The program will display the lowest weight and the average weight. **(Use test pad or back page)*

[7 marks]

6. Write a Python program that processes the monthly sales of 40 salespersons in a company. For each salesperson, the program should ask for their name and monthly sales amount. The program should count how many salespersons achieved the target (sales of 5000 or above) and how many did not achieve the target (sales below 5000). At the end, display the total number of salespersons who achieved the target, the total number who did not, and the average monthly sales. Assume for monthly sales, the user will only enter values ranging from 0 - 20000. **(Use test pad or back page)*

[10 marks]

END OF QUESTION PAPER